CSCI 21023 - Data Communication and Networks

Assignment:
Socket Programming Project

Instructions:

- Students must form **groups of 4 members**.
- The task is to develop a **Socket Programming application** using any preferred programming language (e.g., Python, Java, C/C++, etc.).
- The setup must include **three hosts**:
  - **Two local hosts** (on the same network)
  - **One cloud host** (e.g., AWS, Azure, Google Cloud, or any VPS)
- The application can be based on one or more of the following functionalities:
  - **Chatting system**
  - **File sharing**
  - **Any other communication-based application** using sockets

Technical Requirements:

- Demonstrate proper **client-server communication** between the three hosts.
- Ensure your application handles **basic error checking** and **multiple client connections** (if applicable).
- You may use **TCP or UDP** sockets as appropriate.
- Clearly label IP addresses and ports used in your demonstration.

Evaluation & Viva:

- Each group must **present their project** and **explain the architecture and source code** during the viva session.
- During evaluation, you must **demonstrate your packet transfers using Wireshark**, showing the data exchange between the local and cloud hosts.
- Be prepared to show a **live demonstration** of message or file transfer across all three hosts.

Submission & Deadline:

- **Deadline:** 30th October 2025
- **Submit:**
    - Source code (zipped folder) and Upload to your GitHub Repo
    - A brief report (maximum 2 pages) including:
        - Group members and roles
        - System setup and configuration
        - Screenshots of your application and Wireshark captures